# SOEN 6011 — Delivery 2

## Function F7: $x^y$

Yichen Huang
Student ID: 40167688

Concordia University

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# Outline

1. Problem 5: From-Scratch Implementation and GUI
2. Problem 6: GitHub Repository and Documentation
3. Problem 7: Updated Requirements

# Problem 5 — From D1 to D2

## Delivery 1

- Textual user interface
- Integer powers by exponentiation by squaring
- Non-integer powers using library logarithm and exponential functions
- Validation through built-in numerical utilities

## Delivery 2

- Tkinter graphical user interface
- Integer powers still use exponentiation by squaring
- Natural logarithm implemented from scratch
- Exponential function implemented from scratch
- Custom validation and exception handling

## Main Modification

The D1 hybrid algorithm was preserved, but all mathematical operations required for non-integer exponents were reimplemented without using Python power, logarithm, or exponential functions.

# Problem 5 — From-Scratch Constraints

## Permitted Operations

- Input and output operations
- Basic arithmetic and comparisons
- Explicit `while`-loop iteration
- Exception handling
- Tkinter graphical-interface operations
- Input parsing and output formatting outside the numerical core

## Excluded from the Numerical Core

- `pow()`, `math.pow()`, and `**`
- `math.log()`, `math.exp()`, and `math.isfinite()`
- `range()`, `int()`, and `round()`
- `abs()`, `min()`, and `max()`
- External numerical libraries

## Compliance Boundary

The numerical calculation layer uses custom arithmetic, comparisons, and explicit iteration. Input conversion, result formatting, exception handling, and Tkinter remain separate because they support interaction rather than replace the required numerical algorithms.

## Compliance Result

No prohibited built-in or library function is used for power evaluation, logarithm, exponential evaluation, finite-value checking, integer detection, or numerical iteration.

# Problem 5 — Program Architecture

## GUI and Input Layer

- `PowerCalculatorApp`
- Reads base $x$ and exponent $y$
- Handles Calculate, Clear, and Exit
- Displays results and status messages
- Uses `parse_number()`

## Validation and Exceptions

- `is_finite_number()`
- `is_integer_value()`
- Domain and range checks
- Custom exception hierarchy

## Calculation Layer

`calculate_power(base, exponent)`

⇓

Integer-valued exponent?

| Yes | No |
|---|---|
| ⇓ | ⇓ |
| `power_by_squaring()` | `natural_log()` + `exponential()` |

## Numeric Protection

`_checked_multiply()` detects overflow and severe underflow.

# Problem 5 — Integer Exponent Implementation

## Exponentiation by Squaring

- Used when $y$ is integer-valued
- Requires $O(\log |y|)$ iterations
- Uses arithmetic and loop control only
- Does not use `pow()` or `**`

## Supported Cases

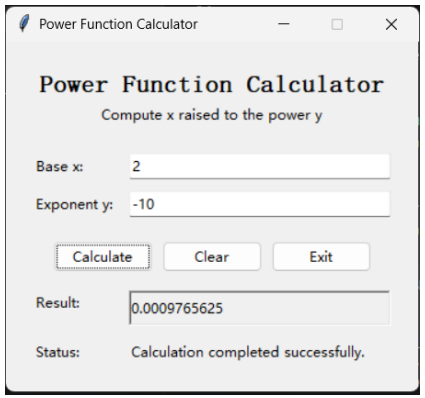
- $y = 0 \Rightarrow x^y = 1$
- $y > 0$: direct integer power
- $y < 0$: invert the base, then use $|y|$
- Negative bases are supported

## Negative Exponent

For $y < 0$, the algorithm first replaces the base with its reciprocal, then applies exponentiation by squaring using $|y|$.

## Core Logic

```
result = 1
factor = base
remaining = exponent
if remaining < 0:
   factor = 1 / factor
   remaining = -remaining
while remaining > 0:
   multiply when odd
   halve remaining
   square factor
```




## Verified Behavior

$2^{-10} = 0.0009765625$

- Correct negative result
- Base inverted first
- No false overflow

# Problem 5 — Tkinter Graphical User Interface

## Interface Components

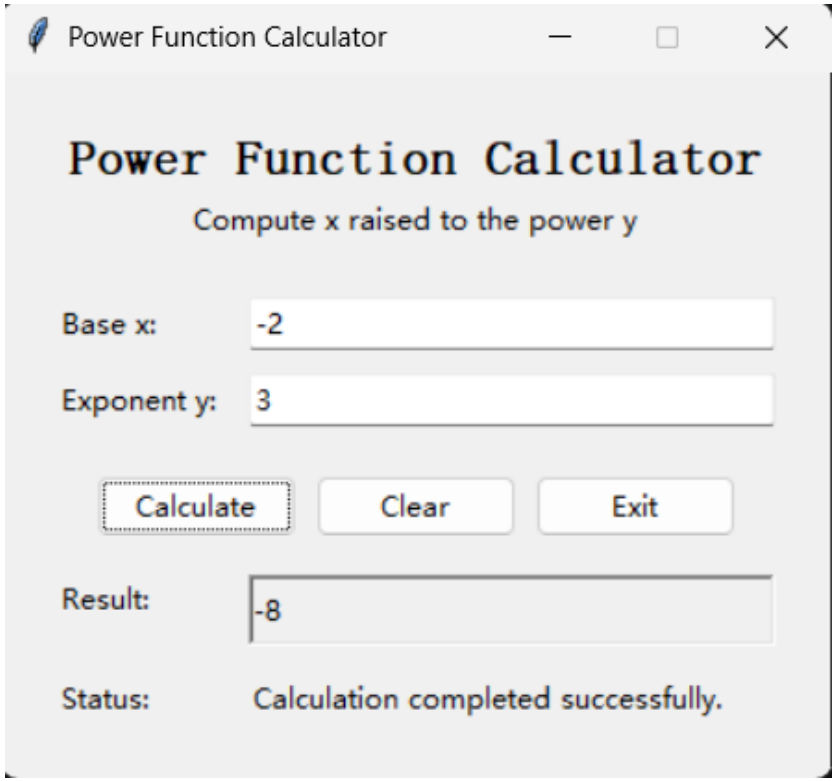
- Labelled fields for base $x$ and exponent $y$
- Calculate, Clear, and Exit buttons
- Separate result and status areas
- Enter key triggers calculation

## Button Behaviour

- **Calculate**: validates and computes
- **Clear**: resets all fields
- **Exit**: closes the application

## Usability Goal

The interface remains open after invalid input so the user can correct the values and try again.




Successful GUI calculation: $(-2)^3 = -8$

## Implementation

The class `PowerCalculatorApp` manages the widgets, event handling, result formatting, and status messages.

# Problem 5 — Exception and Error Handling

## Custom Exceptions

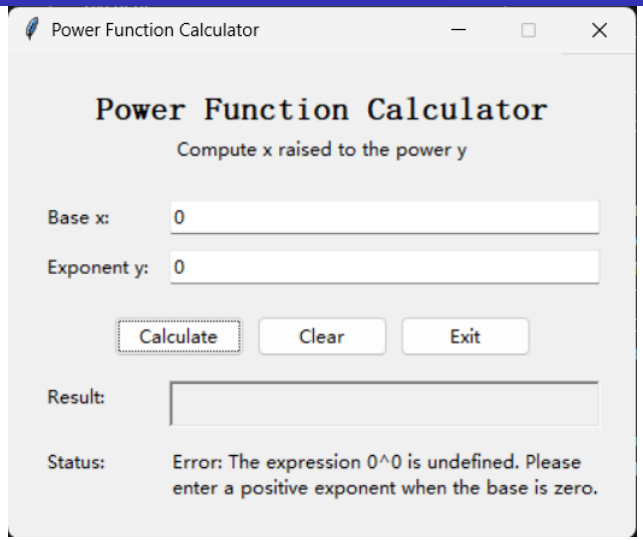
- `InvalidInputError`
- `UnsupportedDomainError`
- `NumericRangeError`
- `ConvergenceError`

## GUI Recovery

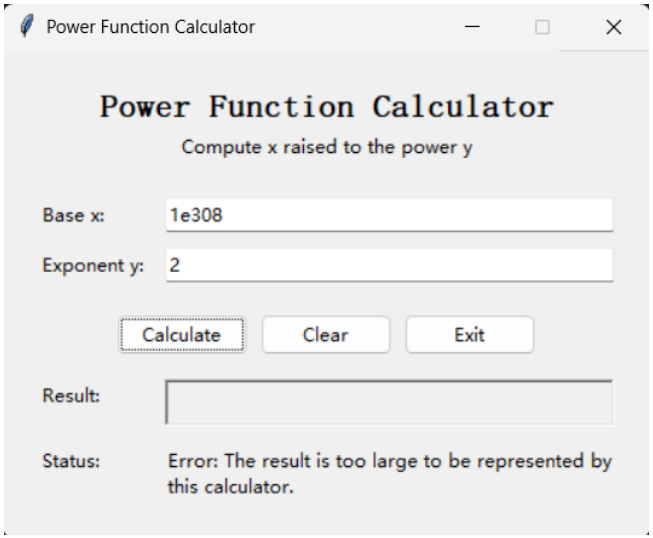
- Catches expected calculator errors
- Clears the result field
- Displays a plain-language message
- Keeps the application open

## User Feedback

Messages explain the error and how the input should be corrected.




Domain validation: $0^0$ is rejected with guidance.




Numeric-range validation: overflow is detected and reported.

# Problem 5 — Verification Results

| Base $x$ | Exponent $y$ | Observed Result | Status |
|---|---|---|---|
| 2 | $-10$ | Result: 0.0009765625 | Passed |
| $-2$ | $-3$ | Result: $-0.125$ | Passed |
| 2 | 0.3 | Result: 1.23114441334492 | Passed |
| 10 | $-400$ | Severe underflow detected: result too small | Passed |
| 10 | 400 | Overflow detected: result too large | Passed |
| 0 | 0 | Rejected: $0^0$ is undefined | Passed |
| $-2$ | 0.5 | Rejected: integer exponent required | Passed |
| abc | 2 | Rejected: numeric base required | Passed |

## Independent Accuracy Check

For $2^{0.3}$, the calculator returned 1.23114441334492, compared with the independently verified expected value 1.2311444133449163. The relative error is below $10^{-9}$.

## Verification Summary

The corrected negative-exponent path produced the expected results. Very small negative-exponent results were classified as underflow, large positive-exponent results were classified as overflow, and the GUI remained open after all expected errors.

# Problem 6 — GitHub Repository and README

## Public Repository

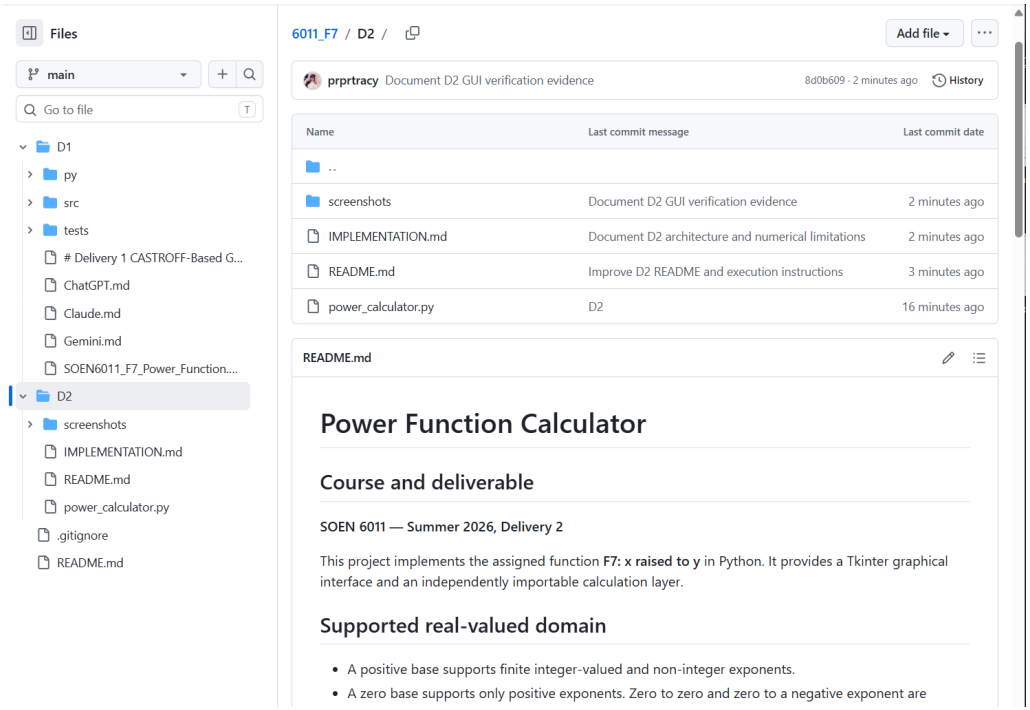
- Hosted using GitHub
- Publicly accessible
- D1 and D2 are stored separately
- Source code, screenshots, and README are included

## README Contents

- Supported mathematical domain
- From-scratch algorithms
- Execution instructions
- Exception handling
- Test cases and limitations

## Repository Address

`https://github.com/prprtracy/6011_F7.git`




Public repository containing the D2 source code, screenshots, and README.

## High-Quality Commit Messages

- `Implement from-scratch hybrid power evaluation`
- `Add domain validation and custom exceptions`
- `Document D2 algorithms, tests, and limitations`

# Problem 7 — Updated Functional and Input Requirements

| ID | Updated Requirement | Change |
|---|---|---|
| FR-001 | The system shall compute $x^y$ within the supported real-valued domain using from-scratch numerical algorithms. | Modified |
| IR-001 | The system shall accept the base $x$ and exponent $y$ through separately labelled input fields. | Modified |
| IR-002 | The system shall accept only finite numeric values for $x$ and $y$. | Retained |
| IR-003 | The system shall determine whether $y$ is integer-valued without using prohibited numerical-library functions. | Modified |
| IR-004 | The system shall allow the user to correct invalid input and perform another calculation without restarting the application. | Modified |
| CR-001 | The system shall not use built-in or library power, logarithm, or exponential operations to evaluate $x^y$. | New |

## D1-to-D2 Modification

The supported mathematical domain remains consistent with D1. The principal changes are the Tkinter input mechanism, custom numerical validation, and the from-scratch implementation of the subordinate logarithm and exponential functions.

# Problem 7 — Updated Output, Error, and Usability Requirements

| ID | Updated Requirement | Change |
|---|---|---|
| OR-001 | The system shall display the computed result in a clearly labelled result area. | Modified |
| ER-001 | The system shall reject $x = 0$ when $y \leq 0$ and display a helpful error message. | Retained |
| ER-002 | The system shall reject $x < 0$ when $y$ is not integer-valued and explain the supported real-valued domain. | Retained |
| ER-003 | The system shall detect overflow, severe underflow, and convergence failure and display a helpful error message. | Modified |
| ER-004 | The system shall recover from invalid input without closing the GUI. | Modified |
| ER-005 | The system shall handle expected input, domain, numeric-range, and convergence errors using specific exception classes without hiding unexpected programming errors. | New |
| UR-001 | The system shall use plain-language labels, status messages, and error messages that explain how the user can correct the input. | Modified |
| UR-002 | The system shall provide a Tkinter graphical user interface for input and output. | Modified |
| UR-003 | The system shall provide Calculate, Clear, and Exit controls and support calculation through the Enter key. | Modified |
| AR-001 | For predefined independently verified test cases, the computed result shall have an absolute error of at most $10^{-9}$ when the expected result is zero, and a relative error of at most $10^{-9}$ otherwise. | Retained |

## Traceability

Each requirement is linked to a GUI element, validation rule, calculation function, exception class, or verification case implemented in Problem 5.

# GAI Use in Delivery 2

| Part | Prompt Type | GAI Contribution | Student Review |
|---|---|---|---|
| P5 | Code generation and technical review | ChatGPT proposed the D1-to-D2 architecture, Tkinter GUI, custom exceptions, and from-scratch numerical methods. Claude provided a second review of the calculation layer and numerical behavior. | Inspected the production code, checked the supported domain, searched for prohibited mathematical operations, ran the GUI, and verified the displayed results. |
| P6 | Repository organization and documentation | Suggested the D2 directory structure, README contents, implementation documentation, screenshot organization, and meaningful commit-message wording. | Reviewed the actual changed files, captured repository evidence, confirmed the public repository, and verified that commit messages describe real changes. |
| P7 | Requirements revision | Drafted ISO/IEC/IEEE 29148-style updated requirements and identified which D1 requirements should be retained, modified, or added. | Compared each requirement with the final D2 code, GUI behavior, exception handling, numerical safeguards, and verification evidence. |

## Use and Non-Use of GAI

GAI was used for drafting, comparison, organization, and refinement. It was not treated as the final authority or used as the sole numerical oracle. Final claims were checked against the actual source code, GUI execution, screenshots, GitHub repository, and project requirements.

# CASTROFF Prompt and Student Responsibility

## CASTROFF-Based Prompt Structure

**Context:** SOEN 6011 Delivery 2 for function $x^y$, extending the completed D1 hybrid implementation.

**Audience:** The professor, teaching assistant, software-engineering students, and the D1 engineering-student persona.

**Scope:** Problems 5, 6, and 7 only.

**Task:** Implement the numerical operations from scratch, create the Tkinter GUI, organize and document the GitHub repository, verify the implementation, and revise the D1 requirements.

**Restrictions:** Do not use built-in mathematical power, logarithm, or exponential operations to evaluate $x^y$. Do not claim that code, screenshots, commits, or repository changes were verified unless actual evidence exists.

**Output:** Working Python code, GUI evidence, repository documentation, meaningful commit history, and a complete updated requirements list.

**Final Review:** Compare every claim with the project description, source code, screenshots, README, and GitHub history.

## Student Responsibility

The GAI outputs were treated as drafts rather than authoritative results. The student reviewed and verified the final code, GUI behavior, supported domain, numerical safeguards, screenshots, repository contents, commit history, requirements, and presentation material.

**Repository Evidence:** The original CASTROFF prompt and GAI outputs are stored as D2/D2_CASTROFF_Prompt.md,

# References


Concordia University,
*SOEN 6011 Project Description*,
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https://github.com/prprtracy/6011_F7.git.

# Thank You

Questions are welcome.

Yichen Huang
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